

Percentages/Everyday Math

Worksheet 4

1. W16/qp41/q1(b,c)

- a. Jakob invests \$500 at a rate of 2% per year compound interest. Claudia invests \$500 at a rate of 2.5% per year simple interest. Calculate the difference between these two investments after 30 years. Give your answer in dollars correct to the nearest cent. [6]
- b. Michel invests \$P at a rate of 3.8% per year compound interest. After 30 years the value of this investment is \$1469. Calculate the value of P. [3]

2. S17/qp41/q1(a,b,c)

Electricity price	Gas price
23.15 cents per day plus 13.5 cents for each unit used	24.5 cents per day plus 5.5 cents for each unit used

- a. An energy company charged these prices in 2013.
- i. In 90 days, the Siddique family used 1885 units of electricity. Calculate the total cost, in dollars, of the electricity they used. [2]
- ii. In 90 days, the gas used by the Khan family cost \$198.16. Calculate the number of units of gas used. [3]
- b. In 2013, the price for each unit of electricity was 13.5 cents. Over the next 3 years, this price increased exponentially at a rate of 8% per year. Calculate the price for each unit of electricity after 3 years. [2]
- c. Over these 3 years, the price for each unit of gas increased from 5.5 cents to 7.7 cents.
- i. Calculate the percentage increase from 5.5 cents to 7.7 cents. [3]
- ii. Over the 3 years, the 5.5 cents increased exponentially by the same percentage each year to 7.7 cents. Calculate the percentage increase each year. [3]

3. W18/qp43/q2 (b and c)

- a. The population of a city is increasing exponentially at a rate of 2% each year. The population now is 256000. Calculate the population after 30 years. Give your answer correct to the nearest thousand. [3]
- b. A bacteria population increases exponentially at a rate of $r\%$ each day. After 32 days, the population has increased by 309%. Find the value of r . [3]

4. M20/qp22/q12

The population of a town decreases exponentially at a rate of 1.7% per year. The population now is 250000. Calculate the population at the end of 5 years. Give your answer correct to the nearest hundred. [3]

5. M20/qp42/q1(f)

The value of Dhanu's model railway is \$550. This value increases exponentially at a rate of $r\%$ per year. At the end of 5 years the value will be \$736. Calculate the value of r . [3]

6. M21/qp22/q13

The population of one variety of butterfly is decreasing exponentially at a rate of 34% per year. At the end of 2014, the population was 125.9 million. Calculate the population at the end of 2019. [2]

Answer Key – Worksheet 4

1. (a)30.68 (b)480 or 479.8
2. (a)(i)275.31 (ii)3203 (b)17 (c)(i)40 (ii)11.9 or 11.86
3. (a)46400 (b)4.5
4. 229500
5. 5.99 or 6
6. 15.8 or 15.76